

graph-theory-symbol— A set of LaTeX commands to have symbols designed for Graph Theory

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? ?

Abstract

This documentation is for package GraphTheorySymbol.

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1 Usage

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call. `{\commandname}` The name of the command to create. `{\commandcode}` The code of the command to create. This macro does nothing. It is merely an example. If this were a real macro, you would put a paragraph here describing what the macro is supposed to do, what its mandatory and optional arguments are, and so forth. Some text about an example macro called `\examplemacro`, which might have an optional argument `[\arg1]` and mandatory one `{\arg2}`.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”. `{\color}` The color that must be used by GraphTheorySymbol afterward.

`\GTSadjacency` This macro will draw an adjacency symbol. A first drawing didn’t include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as `Adj(.,.)`.

2 Implementation

```
1 \package
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{fdsymbol}
6 \RequirePackage{lua-unicode-math}
7 \RequirePackage{xcolor}
8
9 \def \GTS@mainColor {black}
10 \def \GTS@secondColor {gray}
11 \def \GTS@mainColorBackup {black}
12 \def \GTS@secondColorBackup {gray}
13 \def \GTS@drawSlash {}
14 \def \GTS@drawUpArrow {}
```

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```

15 \def \GTS@drawDownArrow {}
16
\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.  $\langle commandname \rangle$  The name of the command to
create.  $\langle commandcode \rangle$  The code of the command to create.
17 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
18   \@ifdefinable{#1}{\def#1/{#2}}
19 }}

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value
"black".  $\langle color \rangle$  The color that must be used by GraphTheorySymbol afterward.
20 \newcommand{\GTSsetMainColor}[1]{%
21 \edef\GTS@mainColorTemp{#1}%
22 \edef\GTS@mainColorBackup{\GTS@mainColor}%
23 \edef\GTS@mainColor{\GTS@mainColorTemp}%
24 }

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial
value "gray".  $\langle color \rangle$  The color that must be used by GraphTheorySymbol afterward.
25 \newcommand{\GTSsetSecondColor}[1]{%
26 \edef\GTS@secondColorTemp{#1}%
27 \edef\GTS@secondColorBackup{\GTS@secondColor}%
28 \edef\GTS@secondColor{\GTS@secondColorTemp}%
29 }

\GTSadjacency This macro will draw an adjacency symbol.
\GTSa 30 \GTS@newcommand{\GTSadjacency}{%
31 \text{\ensuremath{%
32 \raisebox{-0.2ex}{\scalebox{1.5}[1.5]{%
33 %
34 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
35 $\mathcolor{\GTS@secondColor}{\circ}$%
36 }}}%
37 \hspace{-0.39ex}%
38 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
39 $\mathcolor{\GTS@secondColor}{\bullet}$%
40 }}}%
41 \hspace{-0.12ex}%
42 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
43 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
44 }}}%
45 \hspace{-0.40ex}%
46 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
47 $\mathcolor{\GTS@secondColor}{\circ}$%
48 }}}%
49 \hspace{-0.39ex}%
50 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
51 $\mathcolor{\GTS@secondColor}{\bullet}$%
52 }}}%
53 \hspace{0.27ex}%
54 \if\GTS@drawUpArrow\empty\else%
55 \hspace{-0.65ex}%
56 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
57 \textcolor{\GTS@mainColor}{\textasciicircum}%
58 }}}%
59 \fi%
60 \if\GTS@drawDownArrow\empty\else%
61 \hspace{-0.65ex}%
62 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
63 \textcolor{\GTS@mainColor}{\textasciicircum}%
64 }}}%
65 \fi%
66 \if\GTS@drawSlash\empty\else%
67 \hspace{-0.65ex}%
68 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%

```

```

69 \textcolor{black}{/}%
70 }}%
71 \fi%
72 %
73 }}%
74 }}%
75 }
76
77 \GTS@newcommand{\GTSa}{\GTSadjacency/}

```

`\GTSadjacencyRelation` This macro will draw an adjacency symbol with relation spacing in math mode.

```

\GTSaR 78 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSadjacency/}}
79 \GTS@newcommand{\GTSaR}{\GTSadjacencyRelation/}

```

`\GTSunadjacency` This macro will draw an unadjacency symbol.

```

\GTSu 80 \GTS@newcommand{\GTSunadjacency}{%
81 \def\GTS@drawSlash{1}%
82 \GTSadjacency/%
83 \def\GTS@drawSlash{}%
84 }
85
86 \GTS@newcommand{\GTSu}{\GTSunadjacency/}

```

`\GTSunadjacencyRelation` This macro will draw an unadjacency symbol with relation spacing in math mode.

```

\GTSuR 87 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSunadjacency/}}
88 \GTS@newcommand{\GTSuR}{\GTSunadjacencyRelation/}

```

`\GTSdirectedAdjacencyUp` This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

\GTSdAU 89 \GTS@newcommand{\GTSdirectedAdjacencyUp}{%
90 \def\GTS@drawUpArrow{1}%
91 \GTSadjacency/%
92 \def\GTS@drawUpArrow{}%
93 }
94
95 \GTS@newcommand{\GTSdAU}{\GTSdirectedAdjacencyUp/}

```

`\GTSdirectedAdjacencyUpRelation` This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```

\GTSdAUR 96 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{%
97 \mathrel{\GTSdirectedAdjacencyUp/}%
98 }
99 \GTS@newcommand{\GTSdAUR}{\GTSdirectedAdjacencyUpRelation/}

```

`\GTSnegatedDirectedAdjacencyUp` This macro will draw a negated directed adjacency upward symbol.

```

\GTSnDAU 100 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{%
101 \def\GTS@drawSlash{1}%
102 \GTSdirectedAdjacencyUp/%
103 \def\GTS@drawSlash{}%
104 }
105
106 \GTS@newcommand{\GTSnDAU}{\GTSnegatedDirectedAdjacencyUp/}

```

`\GTSnegatedDirectedAdjacencyUpRelation` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```

\GTSnDAUR 107 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
108 \mathrel{\GTSnegatedDirectedAdjacencyUp/}%
109 }
110 \GTS@newcommand{\GTSnDAUR}{\GTSnegatedDirectedAdjacencyDownRelation/}

```

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

111 \GTS@newcommand{\GTSdirectedAdjacencyDown}{%

```

```

112 \def\GTS@drawDownArrow{1}%
113 \GTSadjacency/%
114 \def\GTS@drawDownArrow{%
115 }
116
117 \GTS@newcommand{\GTSdAD}{\GTSdirectedAdjacencyDown/}

```

`\GTSdirectedAdjacencyDown-` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdADR 118 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{%
119 \mathrel{\GTSdirectedAdjacencyDown/}%
120 }
121 \GTS@newcommand{\GTSdADR}{\GTSdirectedAdjacencyDownRelation/}

```

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol.

```

AdjacencyDown 122 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{%
\GTSnDADR 123 \def\GTS@drawSlash{1}%
124 \GTSdirectedAdjacencyDown/%
125 \def\GTS@drawSlash{%
126 }
127
128 \GTS@newcommand{\GTSnDADR}{\GTSnegatedDirectedAdjacencyDown/}

```

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol with relation spacing
`AdjacencyDownRelation` in math mode.

```

\GTSnDADR 129 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
130 \mathrel{\GTSnegatedDirectedAdjacencyDown/}%
131 }
132 \GTS@newcommand{\GTSnDADR}{\GTSnegatedDirectedAdjacencyDownRelation/}

```

`\GTSincidence` This macro will draw an incidence symbol. The operand on the left of the symbol should be
`\GTSi` considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

133 \GTS@newcommand{\GTSincidence}{%
134 \text{\ensuremath{%
135 \raisebox{0ex}{\scalebox{1.8}[1.8]{%
136 %
137 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
138 $\mathcolor{\GTS@secondColor}{\circ}$%
139 }}}%
140 \hspace{-0.39ex}%
141 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
142 $\mathcolor{\GTS@secondColor}{\bullet}$%
143 }}}%
144 \hspace{-0.12ex}%
145 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
146 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
147 }}}%
148 \hspace{-0.18ex}%
149 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
150 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
151 }}}%
152 \hspace{0.2ex}%
153 \if\GTS@drawUpArrow\empty\else%
154 \hspace{-0.65ex}%
155 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
156 \textcolor{\GTS@secondColor}{\textasciicircum}%
157 }}}%
158 \fi%
159 \if\GTS@drawDownArrow\empty\else%
160 \hspace{-0.65ex}%
161 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
162 \textcolor{\GTS@secondColor}{\textasciicircum}%
163 }}}%
164 \fi%
165 \if\GTS@drawSlash\empty\else%

```

```

166 \hspace{-0.65ex}%
167 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
168 \textcolor{black}{/}%
169 }}%
170 \fi%
171 %
172 }}%
173 }}%
174 }
175
176 \GTS@newcommand{\GTSi}{\GTSincidence/}

\GTSincidenceRelation This macro will draw an incidence symbol with relation spacing in math mode.
\GTSiR 177 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSincidence/}}
178 \GTS@newcommand{\GTSiR}{\GTSincidenceRelation/}

\GTSnegatedIncidence This macro will draw a negated incidence symbol.
\GTSnI 179 \GTS@newcommand{\GTSnegatedIncidence}{%
180 \def\GTS@drawSlash{1}%
181 \GTSincidence/%
182 \def\GTS@drawSlash{}%
183 }
184
185 \GTS@newcommand{\GTSnI}{\GTSnegatedIncidence/}

\GTSnegatedIncidence- This macro will draw a negated incidence symbol with relation spacing in math mode.
Relation 186 \GTS@newcommand{\GTSnegatedIncidenceRelation}{%
\GTSnIR 187 \mathrel{\GTSnegatedIncidence/}%
188 }
189 \GTS@newcommand{\GTSnIR}{\GTSnegatedIncidenceRelation/}

\GTSoutIncidence This macro will draw a directed incidence upward (out(ward) incidence) symbol. The
\GTSoI operand on the left of the symbol should be considered the vertex to which the arc/the
operand on the right of the symbol is incident.
190 \GTS@newcommand{\GTSoutIncidence}{%
191 \def\GTS@drawUpArrow{1}%
192 \GTSincidence/%
193 \def\GTS@drawUpArrow{}%
194 }
195
196 \GTS@newcommand{\GTSoI}{\GTSoutIncidence/}

\GTSoutIncidenceRelation This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation
\GTSoIR spacing in math mode.
197 \GTS@newcommand{\GTSoutIncidenceRelation}{%
198 \mathrel{\GTSoutIncidence/}%
199 }
200 \GTS@newcommand{\GTSoIR}{\GTSoutIncidenceRelation/}

\GTSnegatedOutIncidence This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.
\GTSnOI 201 \GTS@newcommand{\GTSnegatedOutIncidence}{%
202 \def\GTS@drawSlash{1}%
203 \GTSoutIncidence/%
204 \def\GTS@drawSlash{}%
205 }
206
207 \GTS@newcommand{\GTSnOI}{\GTSnegatedOutIncidence/}

\GTSnegatedOutIncidence- This macro will draw a negated directed incidence upward (out(ward) incidence) symbol
Relation with relation spacing in math mode.
\GTSnOIR 208 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{%
209 \mathrel{\GTSnegatedOutIncidence/}%
210 }
211 \GTS@newcommand{\GTSnOIR}{\GTSnegatedOutIncidenceRelation/}

```

`\GTSinIncidence` This macro will draw a directed incidence downward (in(ward) incidence) symbol. The
`\GTSiI` operand on the left of the symbol should be considered the vertex to which the arc/the
operand on the right of the symbol is incident.

```

212 \GTS@newcommand{\GTSinIncidence}{%
213 \def\GTS@drawDownArrow{1}%
214 \GTSincidence/%
215 \def\GTS@drawDownArrow{}%
216 }
217
218 \GTS@newcommand{\GTSiI}{\GTSinIncidence/}
```

`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with re-
`\GTSiIR` lation spacing in math mode.

```

219 \GTS@newcommand{\GTSinIncidenceRelation}{%
220 \mathrel{\GTSinIncidence/}%
221 }
222 \GTS@newcommand{\GTSiIR}{\GTSinIncidenceRelation/}
```

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.
`\GTSnII`

```

223 \GTS@newcommand{\GTSnegatedInIncidence}{%
224 \def\GTS@drawSlash{1}%
225 \GTSinIncidence/%
226 \def\GTS@drawSlash{}%
227 }
228
229 \GTS@newcommand{\GTSnII}{\GTSnegatedInIncidence/}
```

`\GTSnegatedInIncidence-` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol
`Relation` with relation spacing in math mode.
`\GTSnIIR`

```

230 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{%
231 \mathrel{\GTSnegatedInIncidence/}%
232 }
233 \GTS@newcommand{\GTSnIIR}{\GTSnegatedInIncidenceRelation/}
```

`\GTSadjacencyMonochrome` This macro will draw a monochrome adjacency symbol.
`\GTSaM`

```

234 \GTS@newcommand{\GTSadjacencyMonochrome}{%
235 \GTSsetMainColor{black}%
236 \GTSsetSecondColor{black}%
237 \GTSadjacency/%
238 \GTSsetMainColor{\GTS@mainColorBackup}%
239 \GTSsetSecondColor{\GTS@secondColorBackup}%
240 }
241
242 \GTS@newcommand{\GTSaM}{\GTSadjacencyMonochrome/}
```

`\GTSadjacencyMonochrome-` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.
`Relation`
`\GTSaMR`

```

243 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{%
244 \mathrel{\GTSadjacencyMonochrome/}%
245 }
246 \GTS@newcommand{\GTSaMR}{\GTSadjacencyMonochromeRelation/}
```

`\GTSunadjacencyMonochrome` This macro will draw a monochrome unadjacency symbol.
`\GTSuM`

```

247 \GTS@newcommand{\GTSunadjacencyMonochrome}{%
248 \def\GTS@drawSlash{1}%
249 \GTSadjacencyMonochrome/%
250 \def\GTS@drawSlash{}%
251 }
252
253 \GTS@newcommand{\GTSuM}{\GTSunadjacencyMonochrome/}
```

`\GTSunadjacencyMonochrome-` This macro will draw a monochrome unadjacency symbol with relation spacing in math
`Relation` mode.
`\GTSuMR`

```

254 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{%
255 \mathrel{\GTSunadjacencyMonochrome/}%
256 }
257 \GTS@newcommand{\GTSuMR}{\GTSunadjacencyMonochromeRelation/}
```

`\GTSdirectedAdjacencyUp-` This macro will draw a monochrome directed adjacency upward symbol. The operand on
`Monochrome` the left of the symbol should be considered the source of the arc; the operand on the right
`\GTSdAUM` of the symbol should be considered the target of the arc.

```

258 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
259 \def\GTS@drawUpArrow{1}%
260 \GTSadjacencyMonochrome/%
261 \def\GTS@drawUpArrow{}}%
262 }
263
264 \GTS@newcommand{\GTSdAUM}{\GTSdirectedAdjacencyUpMonochrome/}
```

`\GTSdirectedAdjacencyUp-` This macro will draw a monochrome directed adjacency upward symbol with relation spacing
`MonochromeRelation` in math mode.

```

\GTSdAUMR 265 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
266 \mathrel{\GTSdirectedAdjacencyUpMonochrome/}%
267 }
268 \GTS@newcommand{\GTSdAUMR}{\GTSdirectedAdjacencyUpMonochromeRelation/}
```

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency upward symbol.
`AdjacencyUpMonochrome`

```

\GTSnDAUM 269 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
270 \def\GTS@drawSlash{1}%
271 \GTSdirectedAdjacencyUpMonochrome/%
272 \def\GTS@drawSlash{}}%
273 }
274
275 \GTS@newcommand{\GTSnDAUM}{\GTSnegatedDirectedAdjacencyUpMonochrome/}
```

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency upward symbol with relation
`AdjacencyUpMonochrome-` spacing in math mode.

```

Relation 276 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
\GTSnDAUMR 277 \mathrel{\GTSnegatedDirectedAdjacencyUpMonochrome/}%
278 }
279 \GTS@newcommand{\GTSnDAUMR}{%
280 \GTSnegatedDirectedAdjacencyDownMonochromeRelation/%
281 }
```

`\GTSdirectedAdjacencyDown-` This macro will draw a monochrome directed adjacency downward symbol. The operand on
`Monochrome` the left of the symbol should be considered the target of the arc; the operand on the right
`\GTSdADM` of the symbol should be considered the source of the arc.

```

282 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
283 \def\GTS@drawDownArrow{1}%
284 \GTSadjacencyMonochrome/%
285 \def\GTS@drawDownArrow{}}%
286 }
287
288 \GTS@newcommand{\GTSdADM}{\GTSdirectedAdjacencyDownMonochrome/}
```

`\GTSdirectedAdjacencyDown-` This macro will draw a monochrome directed adjacency downward symbol with relation
`MonochromeRelation` spacing in math mode.

```

\GTSdADMR 289 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
290 \mathrel{\GTSdirectedAdjacencyDownMonochrome/}%
291 }
292 \GTS@newcommand{\GTSdADMR}{\GTSdirectedAdjacencyDownMonochromeRelation/}
```

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency downward symbol.
`AdjacencyDownMonochrome`

```

\GTSnDADM 293 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
294 \def\GTS@drawSlash{1}%
295 \GTSdirectedAdjacencyDownMonochrome/%
296 \def\GTS@drawSlash{}}%
297 }
298
299 \GTS@newcommand{\GTSnDADM}{\GTSnegatedDirectedAdjacencyDownMonochrome/}
```

`\GTSnegatedDirectedAdjacencyDownMonochrome-` This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```

Relation 300 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
\GTSnDADMR 301 \mathrel{\GTSnegatedDirectedAdjacencyDownMonochrome/}%
302 }
303 \GTS@newcommand{\GTSnDADMR}{%
304 \GTSnegatedDirectedAdjacencyDownMonochromeRelation/%
305 }

```

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the `\GTSiM` symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

306 \GTS@newcommand{\GTSincidenceMonochrome}{%
307 \GTSsetMainColor{black}%
308 \GTSsetSecondColor{black}%
309 \GTSincidence/%
310 \GTSsetMainColor{\GTS@mainColorBackup}%
311 \GTSsetSecondColor{\GTS@secondColorBackup}%
312 }
313
314 \GTS@newcommand{\GTSiM}{\GTSincidenceMonochrome/}

```

`\GTSincidenceMonochrome-` This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```

Relation 315 \GTS@newcommand{\GTSincidenceMonochromeRelation}{%
\GTSiMR 316 \mathrel{\GTSincidenceMonochrome/}%
317 }
318 \GTS@newcommand{\GTSiMR}{\GTSincidenceMonochromeRelation/}

```

`\GTSnegatedIncidenceMonochrome` This macro will draw a monochrome negated incidence symbol.

```

Monochrome 319 \GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
\GTSnIM 320 \def\GTS@drawSlash{1}%
321 \GTSincidenceMonochrome/%
322 \def\GTS@drawSlash{}%
323 }
324
325 \GTS@newcommand{\GTSnIM}{\GTSnegatedIncidenceMonochrome/}

```

`\GTSnegatedIncidenceMonochromeRelation` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```

\GTSnIMR 326 \GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{%
327 \mathrel{\GTSnegatedIncidenceMonochrome/}%
328 }
329 \GTS@newcommand{\GTSnIMR}{\GTSnegatedIncidenceMonochromeRelation/}

```

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

\GTSoIM 330 \GTS@newcommand{\GTSoutIncidenceMonochrome}{%
331 \def\GTS@drawUpArrow{1}%
332 \GTSincidenceMonochrome/%
333 \def\GTS@drawUpArrow{}%
334 }
335
336 \GTS@newcommand{\GTSoIM}{\GTSoutIncidenceMonochrome/}

```

`\GTSoutIncidenceMonochrome-` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

Relation 337 \GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{%
\GTSoIMR 338 \mathrel{\GTSoutIncidenceMonochrome/}%
339 }
340 \GTS@newcommand{\GTSoIMR}{\GTSoutIncidenceMonochromeRelation/}

```

`\GTSnegatedOutIncidenceMonochrome` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```

\GTSnOIM

```



```

341 \GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
342 \def\GTS@drawSlash{1}%
343 \GTSoutIncidenceMonochrome/%
344 \def\GTS@drawSlash{}}%
345 }
346
347 \GTS@newcommand{\GTSnOIM}{\GTSnegatedOutIncidenceMonochrome/}

```

`\GTSnegatedOutIncidenceMonochromeRelation` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

\GTSnOIMR 348 \GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
349 \mathrel{\GTSnegatedOutIncidenceMonochrome/}%
350 }
351 \GTS@newcommand{\GTSnOIMR}{\GTSnegatedOutIncidenceMonochromeRelation/}

```

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

352 \GTS@newcommand{\GTSinIncidenceMonochrome}{%
353 \def\GTS@drawDownArrow{1}%
354 \GTSinIncidenceMonochrome/%
355 \def\GTS@drawDownArrow{}}%
356 }
357
358 \GTS@newcommand{\GTSiIM}{\GTSinIncidenceMonochrome/}

```

`\GTSinIncidenceMonochromeRelation` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

\GTSiIMR 359 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{%
360 \mathrel{\GTSinIncidenceMonochrome/}%
361 }
362 \GTS@newcommand{\GTSiIMR}{\GTSinIncidenceMonochromeRelation/}

```

`\GTSnegatedInIncidenceMonochrome` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

```

\GTSnIIM 363 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
364 \def\GTS@drawSlash{1}%
365 \GTSinIncidenceMonochrome/%
366 \def\GTS@drawSlash{}}%
367 }
368
369 \GTS@newcommand{\GTSnIIM}{\GTSnegatedInIncidenceMonochrome/}

```

`\GTSnegatedInIncidenceMonochromeRelation` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

\GTSnIIMR 370 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
371 \mathrel{\GTSnegatedInIncidenceMonochrome/}%
372 }
373 \GTS@newcommand{\GTSnIIMR}{\GTSnegatedInIncidenceMonochromeRelation/}

```

`\dummyMacro` This is a dummy macro. If it did anything, we'd describe its implementation here. "

```

374 \newcommand{\dummyMacro}{

```

Don't use % to introduce a code comment within a `macrocode` environment. Instead, you should typeset all of your comments with LaTeX—doing so gives much prettier results. For comments within a macro/environment body, just do an `\end{macrocode}`, include some commentary, and do another `\begin{macrocode}`. It's that simple.

```

375 }

```

```

376 \endinput

```

```

377 \</package>

```

3 Change History

v0.1.0		\dummyMacro: Added a spurious change
General: Initial version	1	log entry to show what a change
v1.0.0		<i>within</i> an environment definition
General: First public release	1	looks like. 9

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doing nothing	1
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